

ERP SaaS Module Boundary Definition

Version ۲٫۰

Status: Architectural Decision Document

State: Consolidated Module Ownership Definition

Project: HamarehERP SaaS Platform

۱. Purpose

این سند مرزهای رسمی تمام Bounded Context های سیستم HamarehERP را مشخص می کند.
اهداف:

- تعیین مالکیت Entity ها
- جلوگیری از ایجاد داده های تکراری
- مشخص کردن مسئولیت هر Module
- تعیین API Ownership
- تعیین Event Ownership
- جلوگیری از Coupling مستقیم

اصل اصلی:

هر قابلیت، هر Entity و هر داده فقط یک مالک دارد.

۲. Global Module Rules

Rule ۱ - Single Ownership

هر Entity فقط در یک Module تعریف می شود.

مثال:

صحیح:

Invoice

Owner:

Accounting

غلط:

Invoice

Accounting

+

Sales

Rule ۲ - No Direct Database Access

هیچ Module اجازه ندارد:

- جدول دیگر را Query کند .
- داده دیگر را Update کند .
- Foreign Key مستقیم ایجاد کند .

Rule ۳ - Allowed Communication

ارتباط فقط:

API Contract

یا

Service Contract

یا

Domain Event

۳. SaaS Platform Modules

۳٫۱ Tenant Management

Responsibility

مدیریت مشتری SaaS

Owned Entities

Tenant

Tenant Profile

Tenant Domain

Tenant Status

Tenant Configuration

Provides

API:

Tenant API

Events:

TenantCreated.v\

TenantActivated.v\

TenantSuspended.v\

Depends On

None

۳,۲ Subscription & Billing

Responsibility

مدیریت مدل درآمد SaaS

Owned Entities

Plan

Plan Feature

Subscription

Invoice

Billing Transaction

Provides

Subscription API

Billing API

Events:

SubscriptionActivated.v\

SubscriptionExpired.v\

InvoiceCreated.v\

Depends:

۳,۳ Payment Gateway

Responsibility

ارتباط با سیستم‌های پرداخت

Owned:

Payment Transaction

Payment Provider

Provides:

Payment API

Events:

PaymentSucceeded.v\

PaymentFailed.v\

Depends:

Billing

۴. Identity & Security Core

۴,۱ Identity Management

Owned:

User

Credential

Profile

Session

Provides:

Authentication API

Events:

UserCreated.v\

UserActivated.v\

PasswordChanged.v\

۴,۲ Authorization

Owned:

Role

Permission

Scope

Access Rule

Provides:

Permission Check API

Rules:

تمام Module ها مصرف کننده این بخش هستند.

۵. ERP Foundation Modules

۵,۱ Organization Management

Owned:

Company

Branch

Department

Organization Tree

Events:

CompanyCreated.v\

BranchCreated.v\

۵/۲ Master Data Management

Responsibility

مدیریت داده‌های عمومی مشترک

Owned:

Currency

Unit

Country

Language

Tax Definition

Important:

Master Data مالک تمام داده‌ها نیست.

داده تخصصی Module ها متعلق به خودشان است.

مثال:

Product:

مالک Inventory

Customer:

مالک Sales

Supplier:

مالک Purchase

۵/۳ Workflow Engine

Owned:

Workflow Definition

Workflow Instance

Approval Step

Approval History

Provides:

Workflow API

Events:

WorkflowCompleted.v\

۵/۴ Document Management

Owned:

Document

Attachment

File Metadata

Provides:

Document API

9. ERP Business Modules

9.1 Accounting

Owned:

Chart Of Account

Journal

Accounting Entry

Ledger

Financial Document

Payment Allocation

Consumes:

InvoiceCreated

PurchaseCompleted

SalesCompleted

Produces:

JournalCreated.v\

٩,٢ Inventory

Owned:

Warehouse

Stock

Inventory Movement

Stock Adjustment

Inventory Transaction

Consumes:

SalesConfirmed

PurchaseCompleted

Produces:

InventoryChanged.v\

٩,٣ Sales

Owned:

Customer

Lead

Opportunity

Sales Order

Sales Invoice Reference

Produces:

SalesConfirmed.v\

Consumes:

InventoryChanged

Accounting Rules

୧,୪ Purchase

Owned:

Supplier

Purchase Request

Purchase Order

Purchase Receipt

Produces:

PurchaseCompleted.v\

୧,୫ Manufacturing

Owned:

Bill Of Material

Production Order

Work Order

Production Transaction

Consumes:

InventoryChanged

Accounting Rules

Produces:

ProductionCompleted.v\

6,6 Human Resource

Owned:

Employee

Attendance

Payroll

HR Document

Depends:

Identity

Organization

6,V Project Management

Owned:
Project

Task

Resource Allocation

Project Cost

Depends:
Identity

Organization

V. Ownership Matrix

Entity Type	Owner
Tenant	Tenant Management
User	Identity
Role	Authorization
Company	Organization
Currency	Master Data
Workflow	Workflow Engine
Document	Document Management
Invoice	Accounting
Stock	Inventory
Sales Order	Sales

Entity Type	Owner
Purchase Order	Purchase
BOM	Manufacturing
Employee	HR
Project	Project Management

۸. Final Rule

هیچ Module جدیدی قبل از مشخص شدن موارد زیر ایجاد نمی‌شود:

۱. مسئولیت
۲. مالک Entity
۳. API Contract
۴. Event Contract
۵. Dependency
۶. Tenant Impact